Table 1 Risk of bias in Cohort Studies using Newcastle-Ottawa scale

Study	From the same population	Assessment of exposure	Outcome not present at start	Comparability	Assessment of prognostic factors	Assessment of outcome	Adequate follow-up	Co-Interventions similar
Chen 2016[19]	Yes	Yes	Yes	Probably yes	Yes	Probably yes	Probably yes	Probably yes
Chu 2015[20]	Yes	Yes	Probably yes	Probably yes	Yes	Probably yes	Yes	Probably yes
Fang 2019[21]	Yes	Yes	Yes	Probably yes	Yes	Probably yes	Yes	Probably yes
Guo 2020[22]	Yes	Yes	Probably yes	Probably yes	Yes	Probably yes	Yes	Probably yes
Hilton 2013[23]	Yes	Yes	Probably yes	Probably yes	Yes	Probably yes	Yes	Probably yes
Kumarakulasinghe 2016[24]	Yes	Yes	Yes	Probably yes	Yes	Probably yes	Yes	Probably yes
Sedano 2018[25]	Yes	Yes	Probably yes	Probably yes	Yes	Probably yes	Yes	Probably yes
Sharma 2019[9]	Probably yes	Yes	Yes	Probably yes	Yes	Probably yes	Yes	Probably yes
Zenke 2016[26]	Yes	Yes	Probably yes	Probably yes	Yes	Probably yes	Yes	Probably yes

Table 2 Risk of bias in Case Control Studies using Newcastle-Ottawa scale

Study	Assessment of exposure	Confident outcome	Cases properly selected	Controls properly selected	Match
Cho 2018[27]	Yes	probably no	Yes	Yes	Probably yes
Han 2020[28]	Yes	probably no	Yes	Yes	Probably yes
Kim 2018[29]	Yes	Yes	Yes	Yes	Probably yes

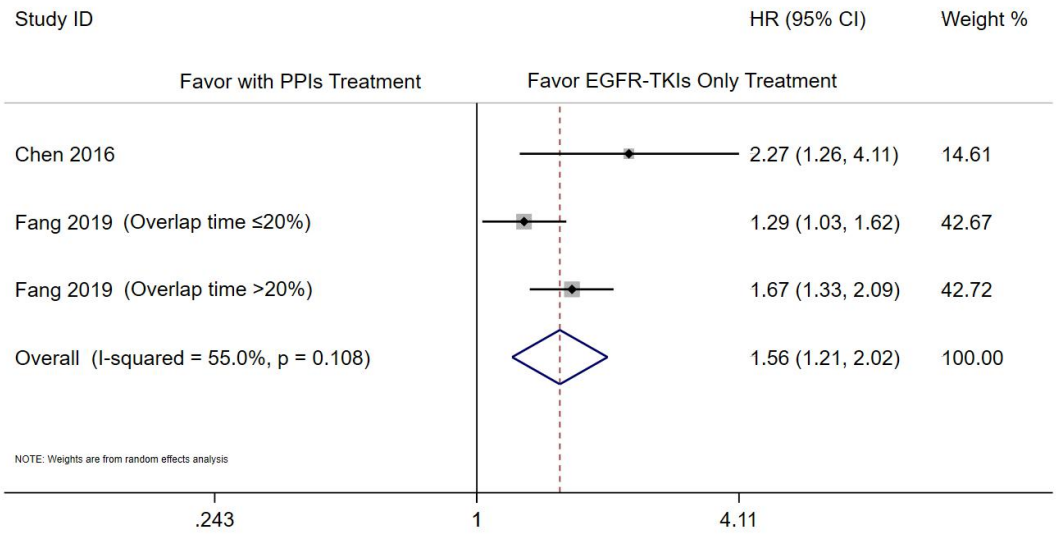

Figure 1 Influence of the concomitant use of PPIs on the OS

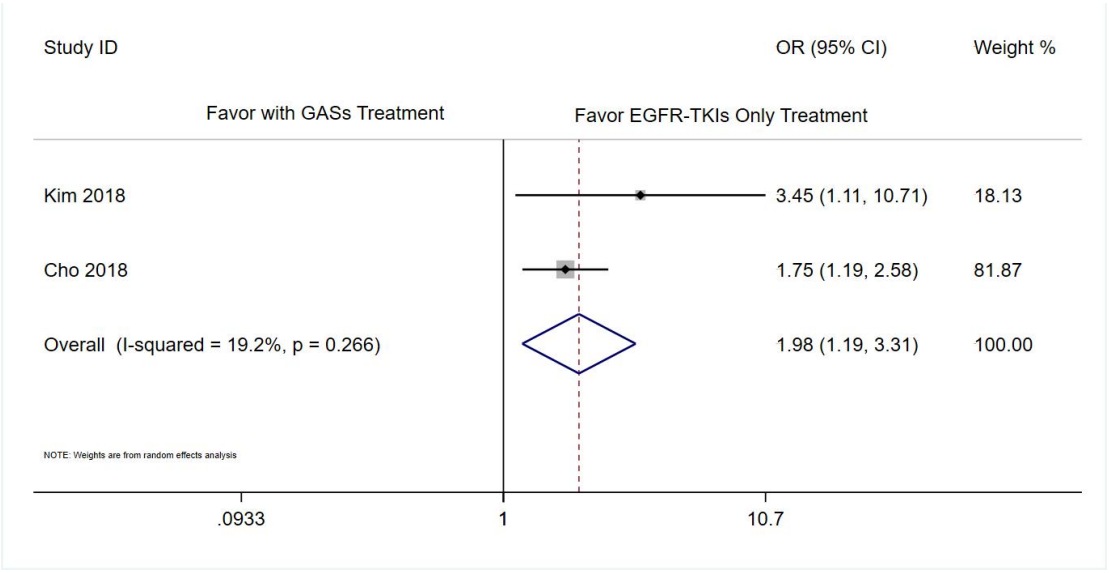

Figure 2 Influence of the use of GASs on the hepatotoxicity

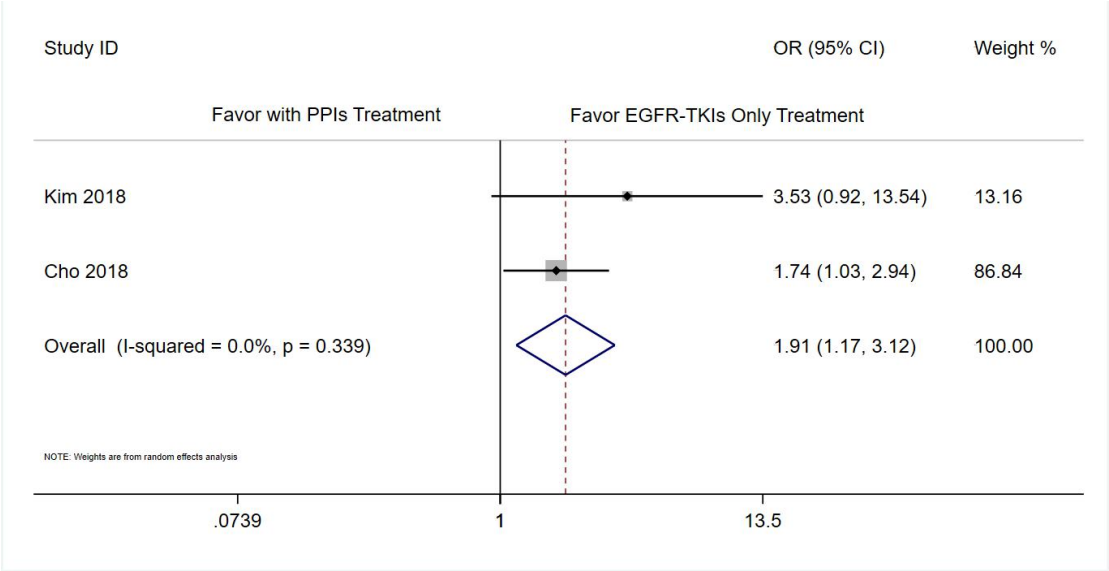

Figure 3 Influence of the use of PPIs on the hepatotoxicity

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